

DIPANKAR NATH

Data Scientist / Decision Science Analyst

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PROFESSIONAL SUMMARY

GenAI Engineer & Data Scientist with **4 years** of experience building and shipping **production LLM, RAG, and agentic-AI systems** in enterprise settings. Owned end-to-end delivery of a production RAG finance assistant, an org-wide AI-native SDLC framework, and a single-handedly-built agentic ServiceNow bot. Deep hands-on Azure AI stack (AI Search, OpenAI, Functions), multi-agent orchestration (Semantic Kernel, AutoGen), and Graph-RAG — turning GenAI into measurable business outcomes across finance, supply chain, and engineering.

TECHNICAL SKILLS

Languages	Python, PySpark, SQL
GenAI / ML	Agentic AI, LLMs, RAG, Graph-RAG, Prompt Engineering, LLM Fine-tuning (LoRA/QLoRA), SLMs (Phi-3), AI-Native SDLC, Multi-Agent Orchestration, Semantic Kernel, AutoGen, LangChain, LlamaIndex, Classification
AI Tooling	Azure AI Search, Azure OpenAI, Azure ML, Azure AI Foundry, GitHub Copilot, Hugging Face, unstructured, Tree-sitter (AST), NVIDIA Nemotron, vector databases
Cloud & Data	Microsoft Azure (AI Search, OpenAI, Durable Functions), Google Cloud (BigQuery), PostgreSQL, SQL
Visualization	Looker, Qlik Sense, Power BI
Databases	PostgreSQL, BigQuery
Dev Tools	Git, Azure Repos / Boards, Jira, FastAPI, Flask

PROFESSIONAL EXPERIENCE

Industrial & Functional Decision Science Analyst

Accenture Strategy & Consulting (Global Network) | 2022 – Present

Flagship GenAI Projects

- RAG Finance Application (Production):** built an enterprise RAG assistant over **1,000–2,000** internal finance documents (reports, contracts, compliance docs) enabling natural-language Q&A and search at **~90%** answer accuracy. Engineered the end-to-end Python pipeline — ingestion → chunking with the unstructured library → embeddings → retrieval — on Azure AI Search + Azure OpenAI. Widely adopted across finance teams in production.
- AI-Native SDLC Framework (Org-wide):** designed a brownfield AI-development framework that builds a reusable **context layer** (domain, code-mapping, patterns, guardrails, hooks), enforces test-driven development, runs **multi-agent code review**, and auto-creates PRs — powered by GitHub Copilot. Rolled out org-wide, accelerating development by **~80%**.
- ServiceNow Agentic Bot:** built an agentic IT-support system on **Semantic Kernel** + Azure OpenAI with a vector-indexed knowledge base and a SQL incident store. The agent interprets tickets, auto-resolves routine issues (password resets, access requests) and escalates the rest — handling **50–60 tickets/week** and saving **30–35 hrs/week** of manual effort.

Additional GenAI & ML Projects

- CodeWiki Bot (Graph-RAG):** converted an entire codebase into a Graph-RAG knowledge graph using AST parsing (Tree-sitter); powers a chatbot, automatic documentation generation, architecture diagram generation, and a bug finder. Standalone, end-to-end.
- SRE Agents (next-gen reliability):** configured and deployed Microsoft's Azure SRE-Agent for specific teams — an autonomous incident-triage system with broad connector support — as the follow-on to the ServiceNow bot. Reduced MTTR by **~80%**. Standalone, end-to-end.
- Legacy Code Converter:** built a sequential **swarm of AutoGen agents** (Azure OpenAI) that converted **~400** legacy .script files to PySpark with auto-generated docs — replacing **~1 month** of manual work. Standalone, end-to-end.
- AI Test-Case Generator:** GenAI tool that generates maximal test cases directly from BRDs, saving **50–60 hrs** of manual test-case writing per cycle. Standalone, end-to-end.
- Synthetic Data + Phi-3 SLM Fine-tuning (Finance):** a two-stage privacy-first pipeline — first generated **100,000+** synthetic financial records at **95%** statistical similarity using NVIDIA Nemotron-4 340B on Azure Durable Functions, then **fine-tuned Microsoft Phi-3-mini (LoRA on Azure ML)** on that data to create a private, domain-expert small language model for finance Q&A — chosen over a large LLM for data privacy, lower training/inference cost, and deep domain knowledge. Deployed to **Azure AI Foundry**.
- Supply-Chain Insights Chatbot (POC):** Azure-OpenAI GPT chatbot for business-data insights with metadata-driven query generation on a Python/FastAPI RAG backend — built as a client-facing proof of concept.

Earlier Data-Science Engagements

- **B2B Lead Generation (Marriott):** analyzed **11M+** small/mid-business records; ran EDA and engineered features for a classification model identifying high-value lead characteristics.
- **CMA Digital Transformation — Saudi Central Bank:** built interactive Qlik Sense dashboards, optimized data scripts, and integrated data sources, translating business requirements into solutions.

EDUCATION

B.Tech NIT Silchar · CGPA 8.1/10	Higher Secondary Board of Secondary Education, Assam · 86%
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CERTIFICATIONS

▪ Microsoft Azure AI Engineer Associate	▪ Machine Learning with Python (IBM)
▪ Microsoft Azure Data Scientist Associate	▪ RAG for Production (ActiveLoop)
▪ Microsoft Azure Fundamentals	▪ LLM Ops Applications (Udacity)
▪ NVIDIA Deep Learning (neural networks & GPU computing)	▪ Google Cloud Digital Leader

KEY COMPETENCIES

ML Expertise: supervised/unsupervised learning, model optimization, feature engineering	Deep Learning: neural networks, transformers, modern AI architectures	Rapid Learning: quickly adopt new tech and adapt to evolving requirements
Analytical Excellence: complex data analysis into actionable insights	Innovation: creative solutions to hard technical challenges	